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Studierichting: Informatica
Oefeningenreeks 5I nr. 2.5

Bereken de booglengte L van de functie $f(x) = \frac{1}{4}(x^2 - 2 \ln x)$ over het interval $[1, e]$.

afgeleide:

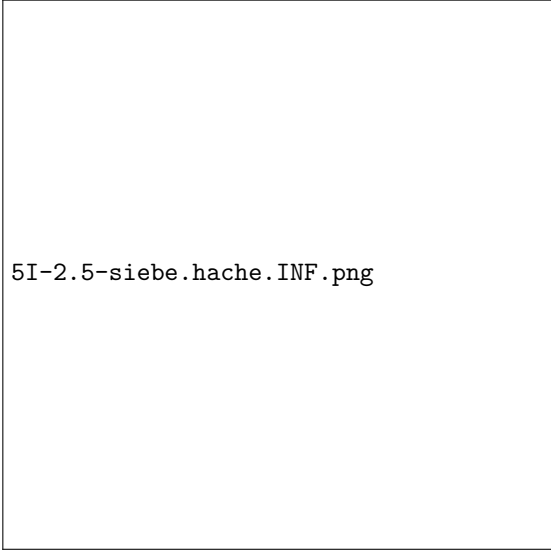
$$f'(x) = \frac{1}{4} \left(2x - \frac{2}{x} \right) = \frac{x}{2} - \frac{1}{2x} = \frac{x^2 - 1}{2x}$$

$1 + (f'(x))^2$:

$$\begin{aligned} 1 + \left(\frac{x^2 - 1}{2x} \right)^2 &= 1 + \frac{x^4 - 2x^2 + 1}{4x^2} \\ &= \frac{4x^2 + x^4 - 2x^2 + 1}{4x^2} = \frac{x^4 + 2x^2 + 1}{4x^2} \\ &= \frac{(x^2 + 1)^2}{(2x)^2} = \left(\frac{x^2 + 1}{2x} \right)^2 \end{aligned}$$

integraal:

$$\begin{aligned} L &= \int_1^e \left(\frac{x}{2} + \frac{1}{2x} \right) dx \\ &= \left[\frac{x^2}{4} + \frac{1}{2} \ln |x| \right]_1^e \\ &= \left(\frac{e^2}{4} + \frac{1}{2} \ln e \right) - \left(\frac{1^2}{4} + \frac{1}{2} \ln 1 \right) \\ &= \left(\frac{e^2}{4} + \frac{1}{2} \right) - \left(\frac{1}{4} + 0 \right) \\ &= \frac{e^2}{4} + \frac{2}{4} - \frac{1}{4} = \frac{e^2 + 1}{4} \end{aligned}$$



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References

- [1] Grafiek gemaakt met Geogebra